RESEARCH & PROTOTYPE REVIEW

Idea Validation Report

Evidence-led review of problem fit, implementation evidence, user-validation gaps and next experiments.

SIH 2026 | PS SIH26250 | Alchemy Architects

SOURCE-CODE REVIEW + PUBLIC RESEARCH | 20 OCTOBER 2026

ALCHEMY ARCHITECTS

Executive summary

AAKAASH ADHINAYAK is a candidate evidence-to-decision resource-planning concept for SIH26250, “Air Power - Dynamic Air Operations & Resource Optimisation.” The product concept combines structured data intake, data-quality checks, deterministic scheduling, independent plan validation, controlled disruption replay, visual status views, an evidence ledger and a VAYU NEXA AI interface.

Validation verdict: PROMISING CONCEPT; FIELD, DATA AND MODEL VALIDATION STILL REQUIRED. The supplied source bundle contains a substantial interface and deterministic scheduling code. It does not, by itself, establish real customer need, authorized operational data access, a trained predictive model, claimed model metrics, or successful end-to-end operation of every external connector.

1. What exactly is being validated?

The hypothesis is that users responsible for complex resource schedules need a single workflow that can reconcile heterogeneous records, expose unknown or stale data, generate a constraint-aware candidate schedule, independently check it, and explain changes after a disruption. The target decision context is safety-sensitive and defence-related; this report treats the implementation as a non-operational student research prototype using generic, public and synthetic data.

2. Evidence actually inspected

- Source bundle includes React/TypeScript modules for Command Centre, Data Quality Studio, Planning Sandbox, Dynamic Replanning, Evidence Ledger, Requirements Coverage, a 3D constellation, weather/public data views, VAYU NEXA, a predictive-maintenance lab and model-evaluation screen.
- The original solver source implements deterministic priority-ranked scheduling, candidate-slot scoring, a greedy first-fit baseline and a separate validation function. These are conventional algorithmic methods, not evidence of machine learning.
- The predictive-maintenance UI and API expose hard-coded baseline/candidate metrics and a short hand-coded C-MAPSS-like sample. A training pipeline or runtime model-inference implementation was not found in the inspected source paths.
- Screenshots show a dark dashboard, planning sandbox, data connector panel and responsive navigation. They are provided export screenshots, not independently recorded live sessions.
- The current curated GitHub demo has its own passing CI checks, but that test result must not be conflated with the original hosted AI Studio app.

3. Idea validation matrix

Validation question	Current evidence	Status	Next evidence required
Does the solution map to the problem statement?	Concept modules directly address data quality, scheduling, disruption replay and auditability.	Plausible	Map every official PS requirement to a tested acceptance case.
Does the source contain real scheduling logic?	Deterministic TypeScript heuristic and independent validator are present.	Code present; edge cases remain	Run reproducible feasible, infeasible, dependency and deadline tests.
Are public data integrations real?	Code attempts public weather fetch; other screens use seed/fallback content.	Partially evidenced	Capture timestamped API responses and failure-mode logs.
Does predictive ML/DL run?	Metrics and predicted values are prefilled in UI/seed code; no training code found.	Not validated	Implement and reproduce an actual training/evaluation pipeline.
Has a stakeholder confirmed the pain point?	No signed interview notes or visit evidence were supplied.	Not validated	Conduct and document authorized stakeholder conversations.
Is the business model validated?	Only plausible user groups and revenue hypotheses are available.	Not validated	Interview users, price-test a bounded civilian/education pilot.

4. Preliminary fit scorecard (analyst judgement, not survey results)

Dimension	Score / 5	Reason
Problem-solution fit	4	Features correspond to plausible data and scheduling pain points, but official PS mapping and users should verify fit.
Software demonstrability	3	Substantial UI and deterministic logic exist in source; the hosted flow was not fully smoke-tested.
Data readiness	2	One public civil-weather connector path exists; other data appears seeded or requires endpoint/licence verification.
ML/DL evidence	1	Predictive results are displayed, but no train/inference/evaluation code was found.
Stakeholder validation	1	No authenticated field notes, interviews, or signed validation letters were supplied.
Security/governance	2	Security/admin concepts exist in UI; original Firestore rules and source bundle need review before any public or real-data deployment.

5. Recommended validation experiments

- Scenario benchmark: create 20-50 clearly synthetic generic tasks and resources, including unknown states, maintenance blocks, duplicate records, dependencies, impossible deadlines and capacity conflicts.
- Compare the heuristic with first-fit baseline on the same dataset/version; measure feasibility, number assigned, tardiness, objective score and wall-clock runtime. Do not label a heuristic result “optimal” without a proof-producing solver or bounded exhaustive comparison.
- Independent validator adversarial tests: manually inject double-booking, wrong capability, out-of-window, stale source and invalid dependency cases and prove rejection.
- Data adapter verification: save request URL, status, retrieval time, source observation time, schema checks, terms/licence and the exact response sample. Distinguish LIVE, CACHED, SYNTHETIC and FAILED states.
- Stakeholder validation: obtain permission before visiting or interviewing; record date, office, role, consent, questions, notes and supporting artefacts. Keep public/civil analogues distinct from defence-specific validation.
- ML experiment: download the exact permitted NASA C-MAPSS asset, publish its source/version and preprocessing, keep engines separated between train/validation/test, establish a baseline, report RMSE/MAE and evaluate uncertainty on held-out units.

6. Go / no-go recommendation

GO for a bounded research prototype and supervised demonstration. NO-GO for claims of operational readiness, proven savings, model accuracy, authenticated defence data integration or adoption by any ministry/office. The strongest next evidence is not another screen: it is a reproducible test pack, verified public connector response, documented stakeholder feedback and an honest feature-status matrix.

Suggested PPT sentence: “AAKAASH ADHINAYAK demonstrates an evidence-first planning workflow with deterministic scheduling and independent validation; field validation and predictive-model benchmarking are the next milestones.”

References and source notes

[1] AI Studio application URL
https://aakaash-adhinayak.ai.studio/
User-supplied hosted prototype link. Automated retrieval was unsuccessful in this review; screenshots and the exported source bundle were inspected instead.

[2] Google AI Studio export README
https://ai.studio/apps/2ca7a5fb-2cbb-44ac-8b9c-15f6e7a7e628
The provided export metadata points to this AI Studio app identifier; used to contextualize export contents.

[3] AviationWeather.gov Data API
https://aviationweather.gov/data/api/
Official public civil aviation weather data API documentation; connector status and terms must be verified before live claims.

[4] NASA Prognostics Center of Excellence Dataset Repository
https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/

Official NASA repository lists turbofan engine degradation simulation data and the C-MAPSS benchmark. It is not military telemetry.